

Chapitre 5. SIFT Descriptor and Applications

1.1 Introduction

1.2 What is an interest point?

1.3 Detecting blobs

1.4 SIFT Detector

1.5 SIFT Descriptor

Chapitre 5. SIFT Descriptor and Applications

1.1 Introduction

From the studied course in semester 1, we know:

- how to find edges and corners in images,
- compute the boundary of objects.

If we try to recognize object with **fairly complex appearances**, we must be able to match **features** that are fairly descriptive and unique.

SIFT feature detector is able to perform this task.

Chapitre 5. SIFT Descriptor and Applications

1.1 Introduction

To recognize objects, we can use some techniques such as:

- apply threshold and
- get binary image and
- compute geometry properties of these objects to recognize them.



Chapitre 5. SIFT Descriptor and Applications

1.1 Introduction

Problem: Find objects in a query image knowing a model image

Solution: We can use template matching.

Problems: When image is rotated and magnified differently.

Solution: We need then different templates of the object under different rotations and scales.

In addition, objects are occluded in the query image.



Chapitre 5. SIFT Descriptor and Applications

1.1 Introduction

Other solution:

Extract directly from image some descriptive unique features.

We will use SIFT feature detector (Scale Invariant Feature Transform) proposed by D. Lowe in 2004.



Chapitre 5. SIFT Descriptor and Applications

1.1 Introduction

We study:

- The theory behind this descriptor
- How to use it to solve some vision problems:
Image alignment,
Images stitching (panoramic images),
2D Object recognition.



Chapitre 5. SIFT Descriptor and Applications

1.1 Introduction

-1.2 What is an interest point?

From the state of the art, edges and corners are not interesting for a lot of applications.

-The **blob**, is an interesting point, which has some local appearance within it useful for matching.



Chapitre 5. SIFT Descriptor and Applications

1.1 Introduction

We study in this chapter:

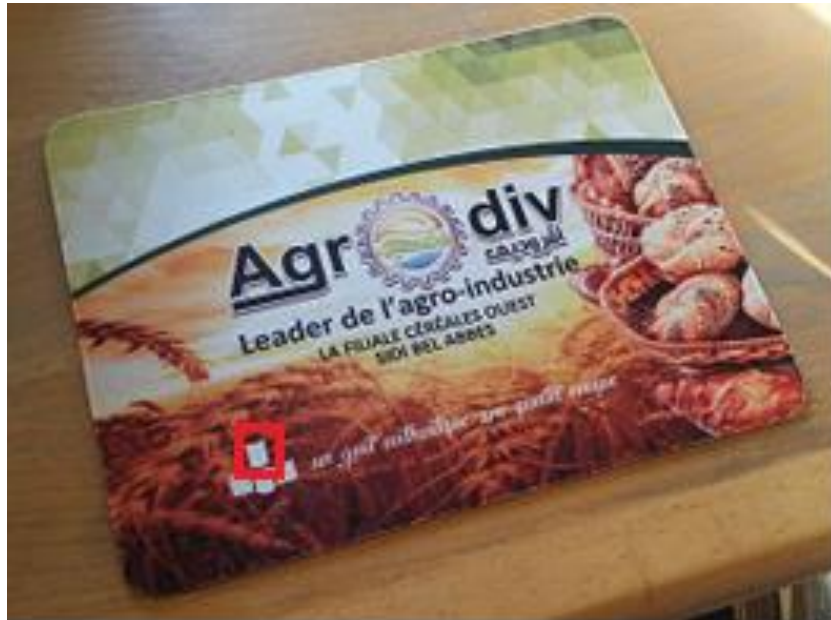
- What is an interest point?
- Detecting a blob: it is a patch with local appearance
- SIFT Detector (David Low)
- SIFT Descriptor (extract a description or signature)



Chapitre 5. SIFT Descriptor and Applications

1.2 What is an interest point?

Two images of the same object may have differences in lighting (distribution), brightness (intensity), orientation, size, etc.



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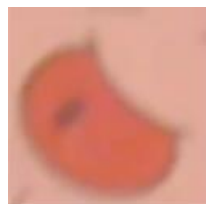
1.2 What is an interest point?

A naïve algorithm for patch matching:

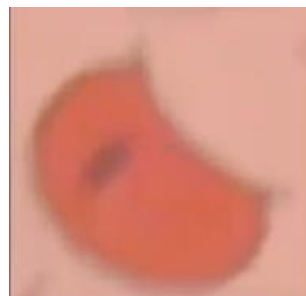
Rescale, rotate (eliminate the differences) the patch query in order to be able to matched with the match model.



Model



Query



Scaled query



Rotated scaled query

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1.2 What is an interest point?

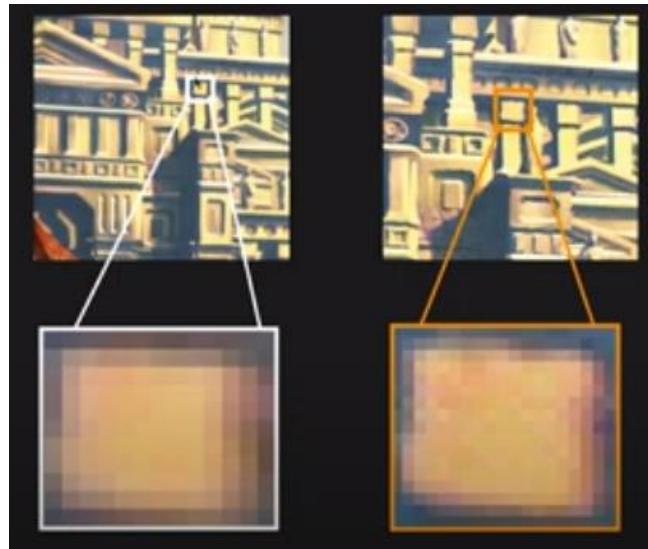
The local appearance around the **interest point** should be rich enough in terms of brightness variations, color variation, etc., that there is a certain amount of uniqueness to it that can be exploited for matching purposes.

- Has well-defined representation (signature) for matching/comparing with other points.
- Has a well-defined position in the image.
- Should be invariant to image rotation and scaling.
- Should be insensitive to lighting changes.

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1.2 What is an interest point?

- Boundaries are not descriptive enough
- Blobs are descriptive enough, potentially good interest points.

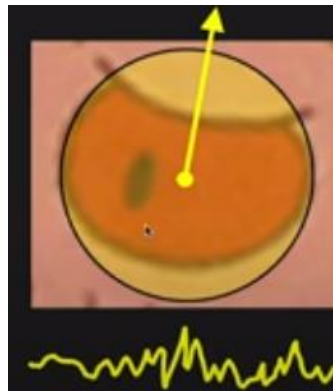


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1.2 What is an interest point?

For a blob-like feature to be useful, we need to:

- Locate the blob
- Determine its size
- Determine its orientation
- Formulate a description or signature that is independent of size and orientation.



Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs

Gaussian filter is used for removing noise by smoothing

A blob is a region of an image in which some properties are constant or approximately constant; all the points in a blob can be considered in some sense to be similar to each other. The most common method for blob detection is convolution.

Given some property of interest expressed as a function of position on the image, there are two main classes of blob detectors:

- (i) *differential methods*, which are based on derivatives of the function with respect to position, and
- (ii) *methods based on local extrema*, which are based on finding the local maxima and minima of the function.

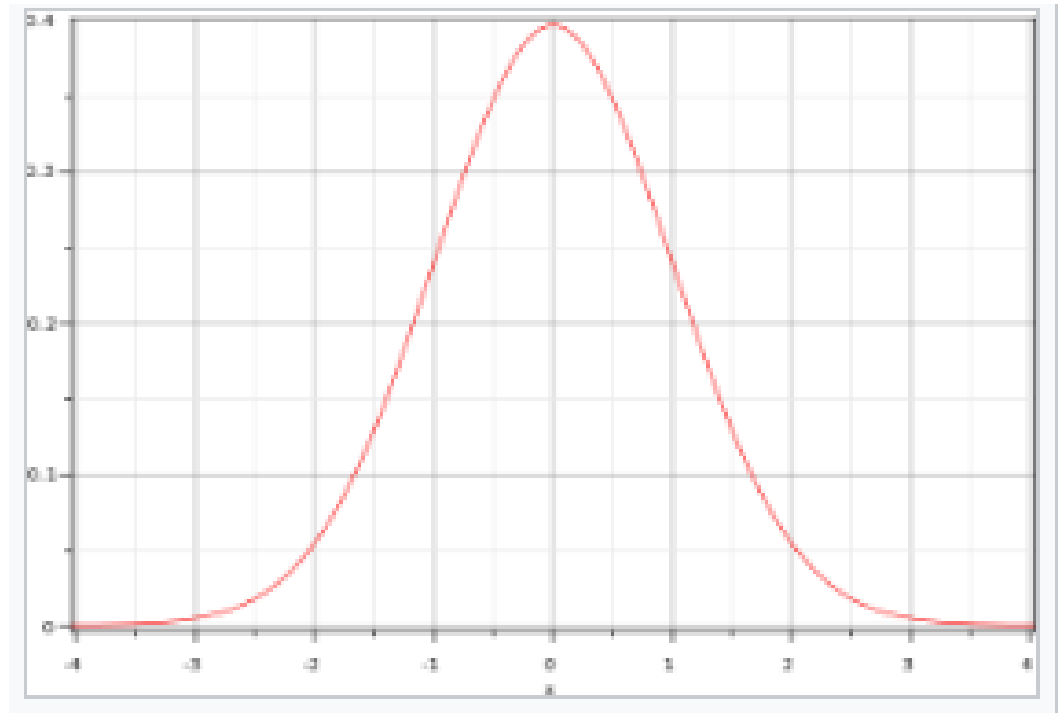
Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs

Gaussian filter is used for removing noise by smoothing

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

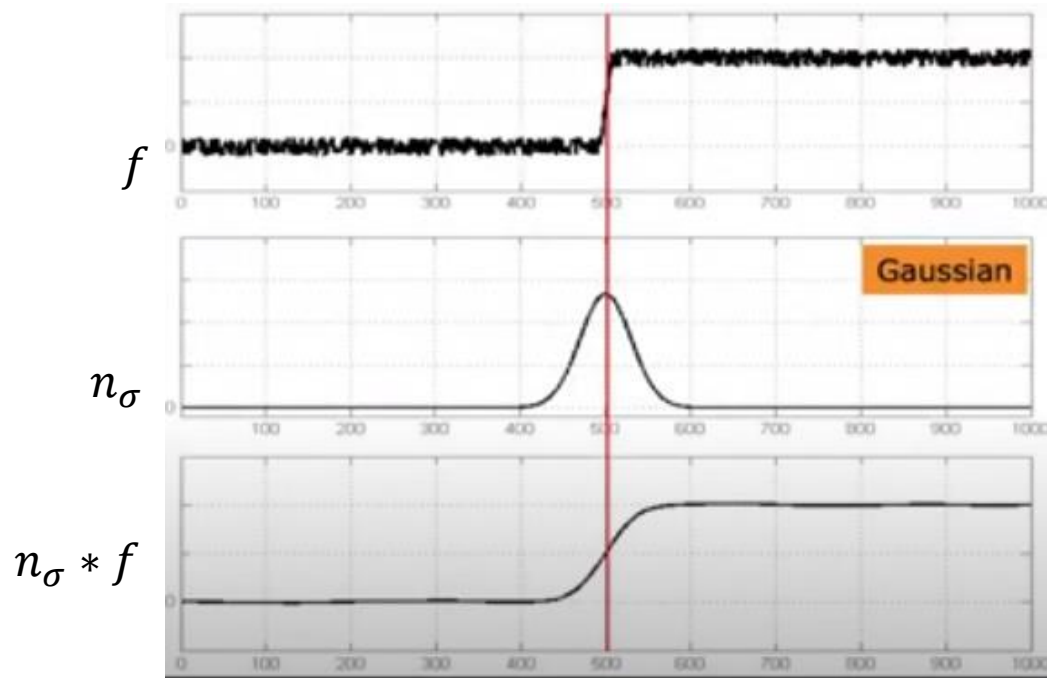
$$g(x, y) = \frac{1}{2\pi\sigma^2} \cdot e^{-\frac{x^2+y^2}{2\sigma^2}}$$



Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs

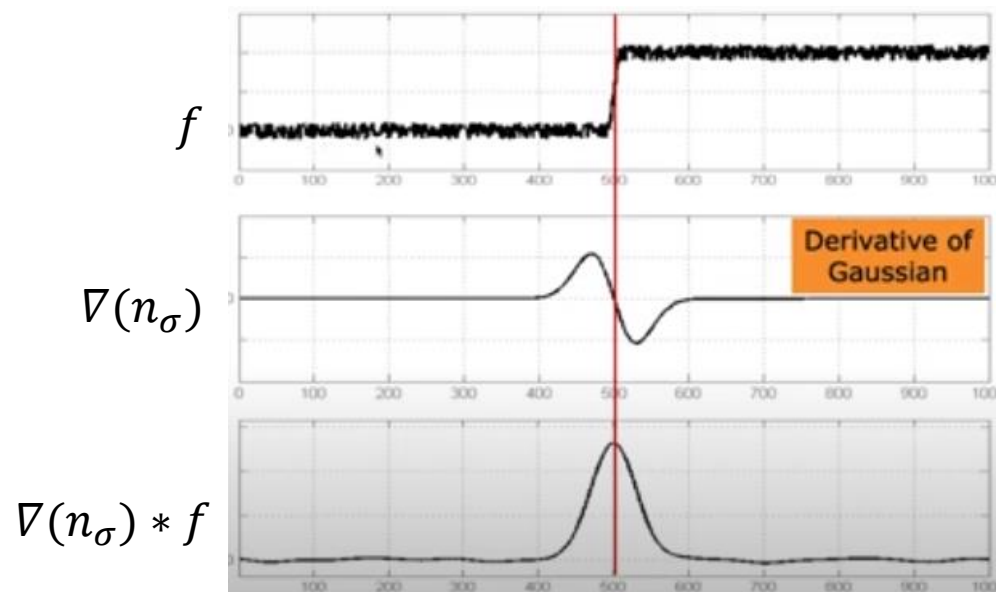
Gaussian filter is used for removing noise by smoothing



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1.3 Detecting blobs

Extremum of Derivative of Gaussian filter denotes an edge.

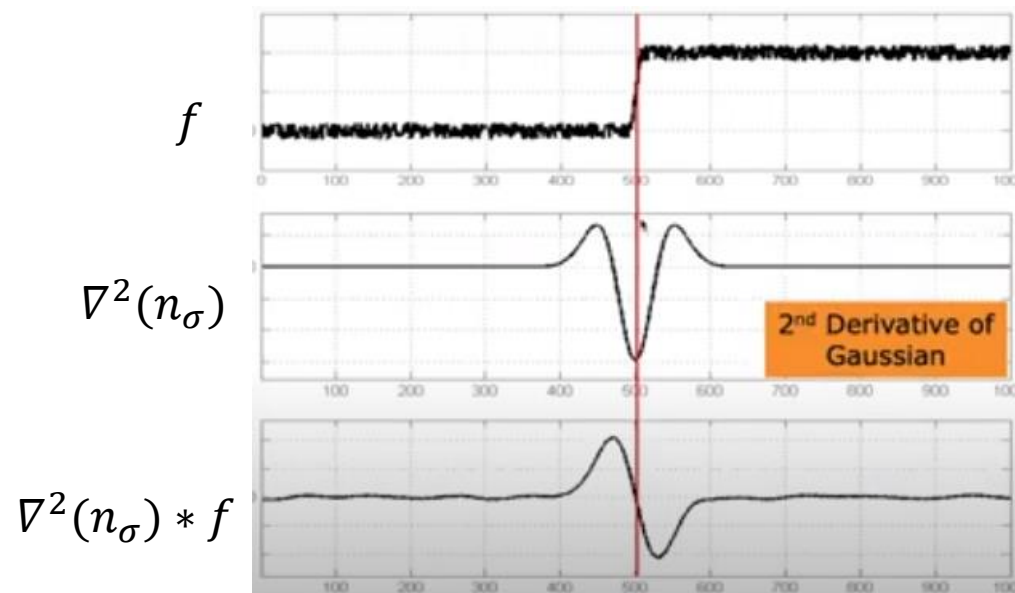


A clear peak at the location of the edge.

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1.3 Detecting blobs

Zero crossing of the second Derivative of Gaussian filter denotes an edge.

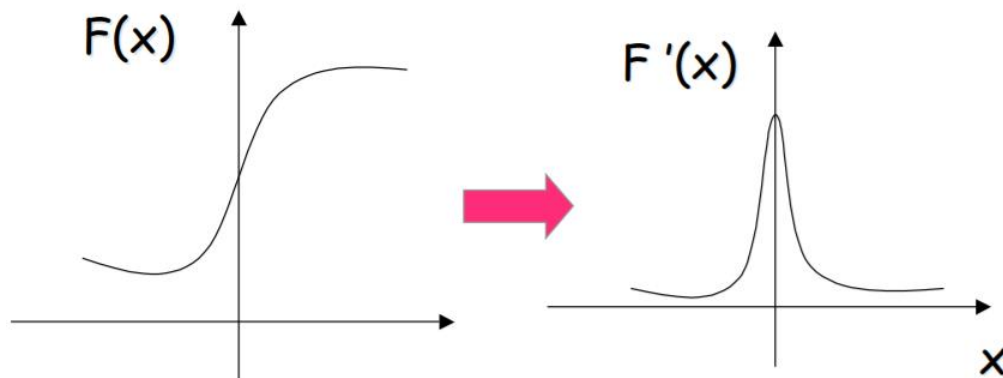
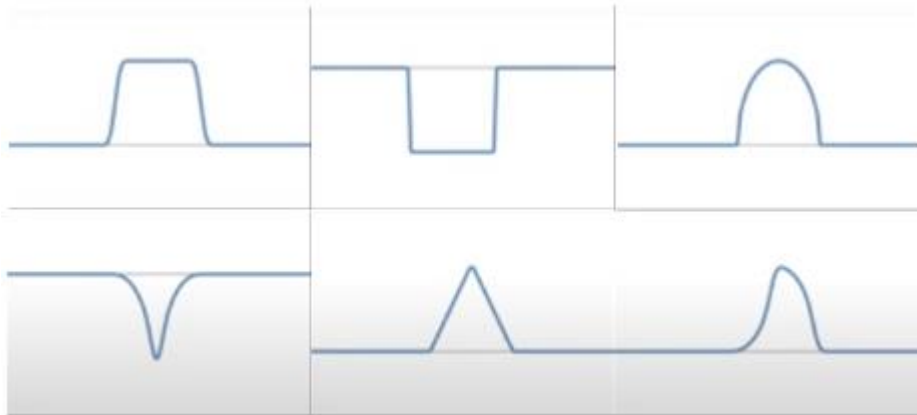


We can use a second derivative of the image and detect the zero crossing which denotes the edge location.

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1.3 Detecting blobs

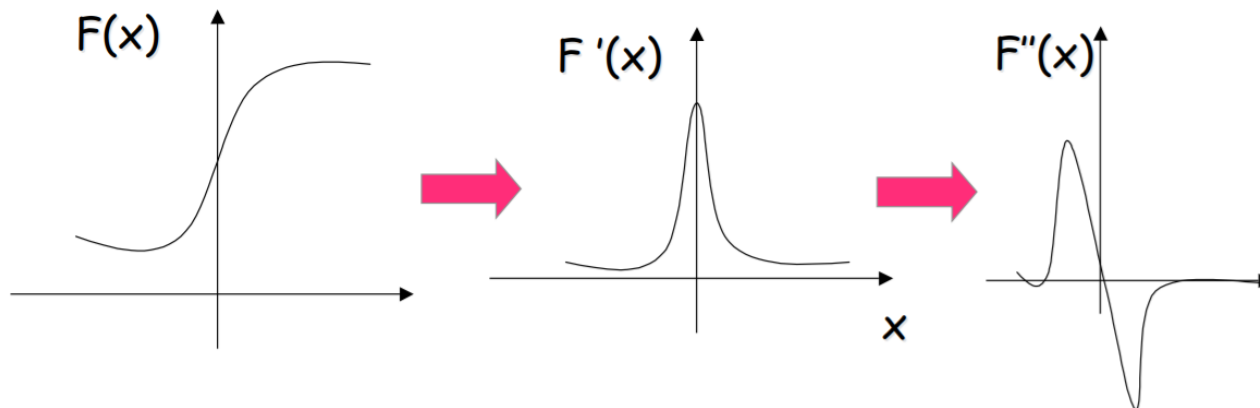
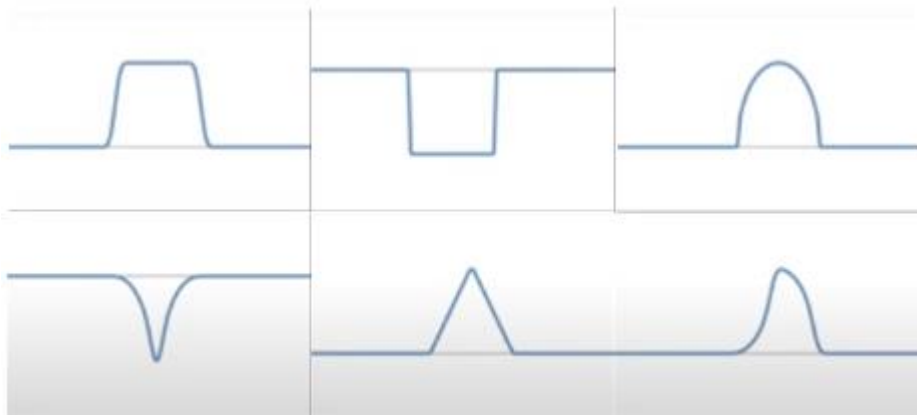
Example of 1D blob-like structures



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1.3 Detecting blobs

Example of 1D blob-like structures



Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs

Convolution of an image with a Gaussian filter

$$\nabla^2(f(x, y) \otimes G(x, y)) = \nabla^2 G(x, y) \otimes f(x, y)$$

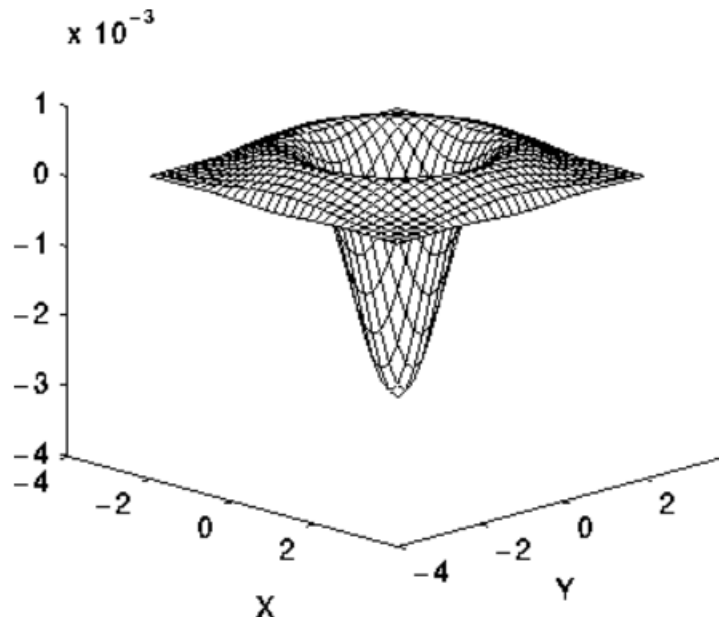
$$G(x, y, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

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1.3 Detecting blobs

Example of 1D blob-like structures

$$G(x, y, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x^2+y^2)}{2\sigma^2}} \quad LoG(x, y) = -\frac{1}{\pi\sigma^4} \left[1 - \frac{x^2 + y^2}{2\sigma^2} \right] e^{-\frac{x^2+y^2}{2\sigma^2}}$$



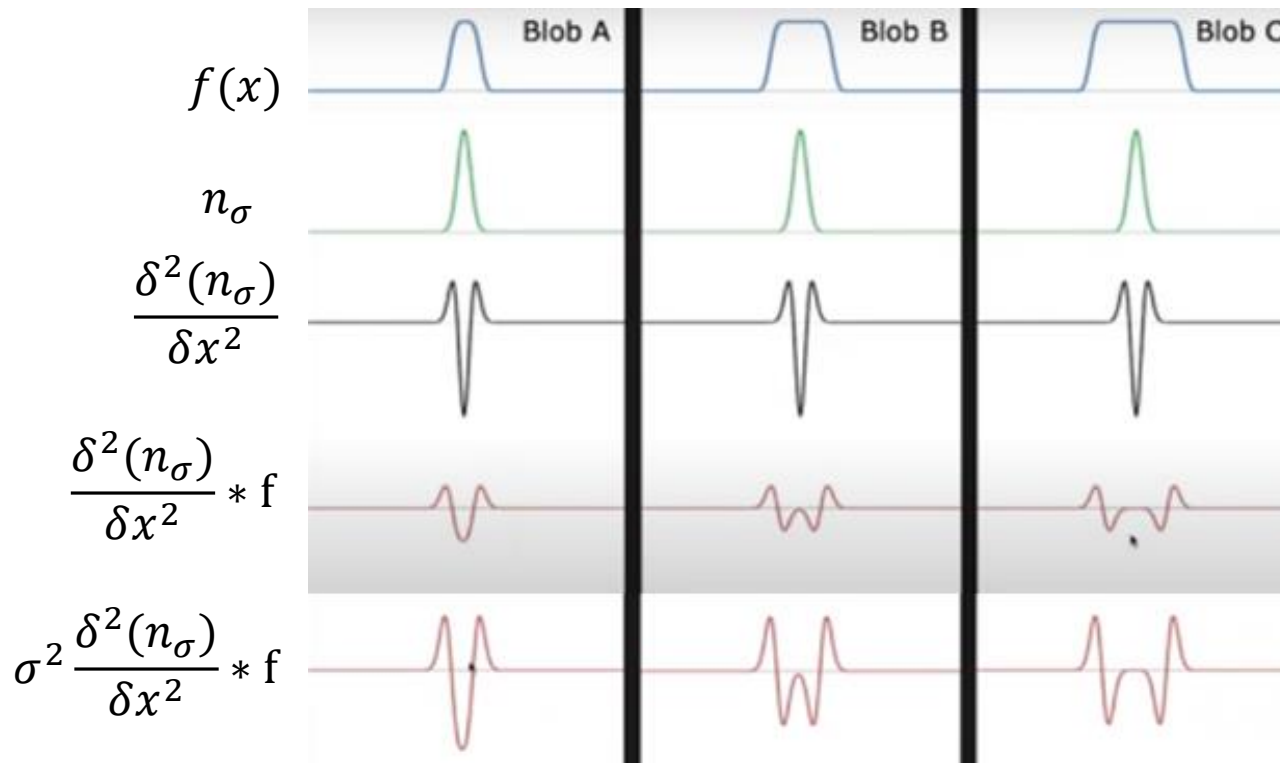
0	1	1	2	2	2	1	1	0
1	2	4	5	5	5	4	2	1
1	4	5	3	0	3	5	4	1
2	5	3	-12	-24	-12	3	5	2
2	5	0	-24	-40	-24	0	5	2
2	5	3	-12	-24	-12	3	5	2
1	4	5	3	0	3	5	4	1
1	2	4	5	5	5	4	2	1
0	1	1	2	2	2	1	1	0

for a Gaussian $\sigma = 1.4$

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1.3 Detecting blobs

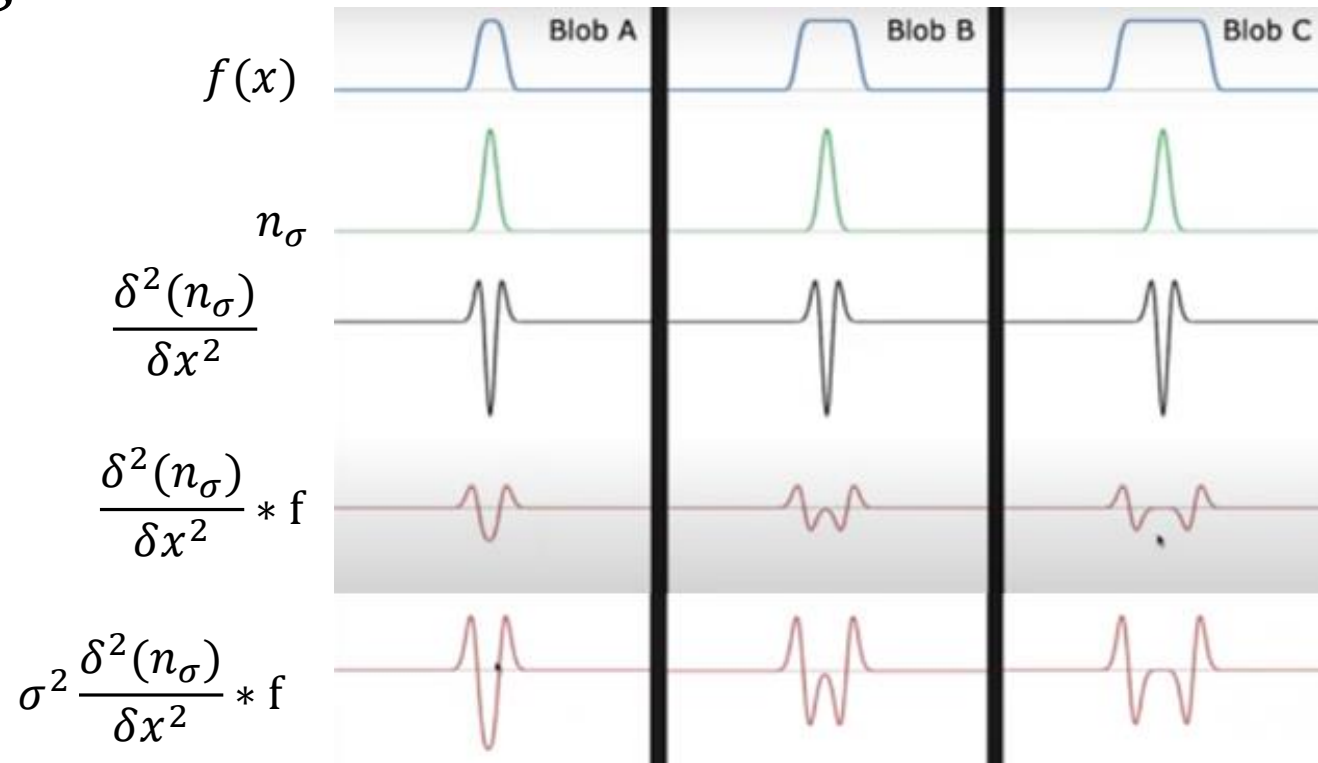
Example of 1D blob-like structures



The response to the operator Derivative of Gaussian depends on the sigma

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1.3 Detecting blobs

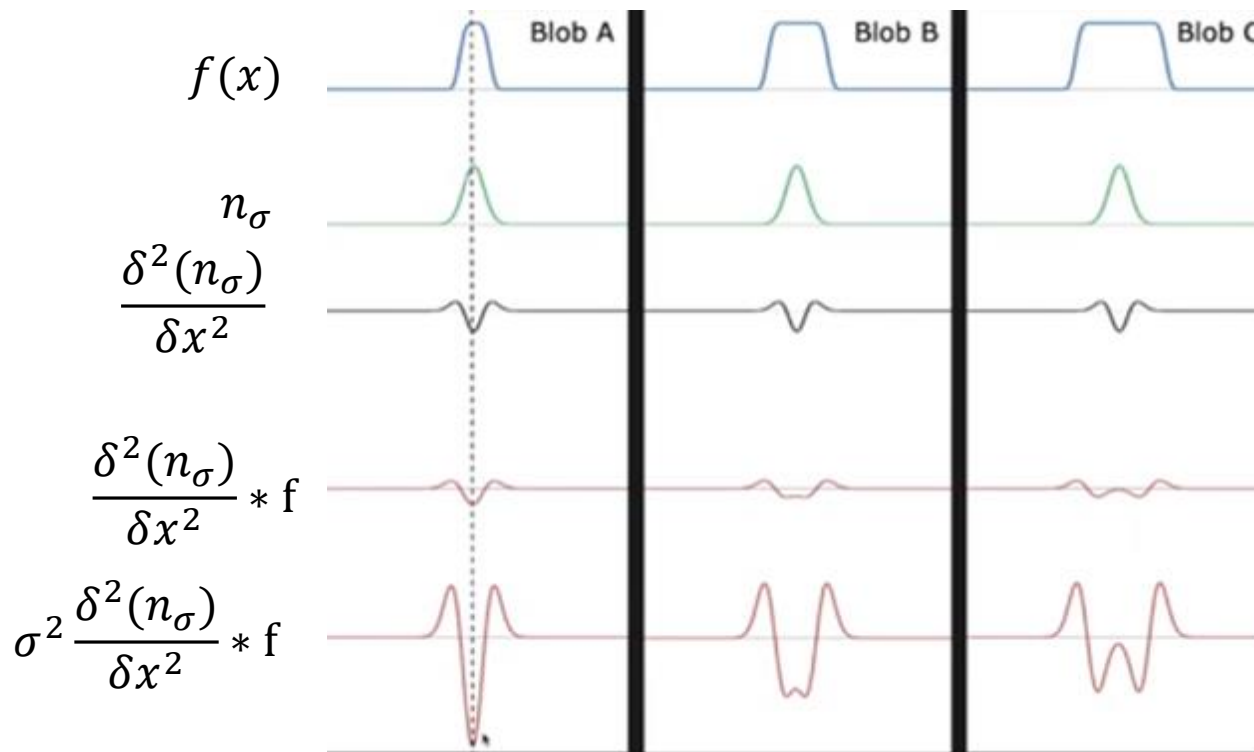


If we apply the second derivative of the Gaussian shown in the third row to the image in the first row, we end up with the output shown in the fourth row. Note that we get two distinct zero-crossings in the output for blob C, but since blobs A and B are thinner, in these cases, the two zero-crossings overlap.

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1.3 Detecting blobs

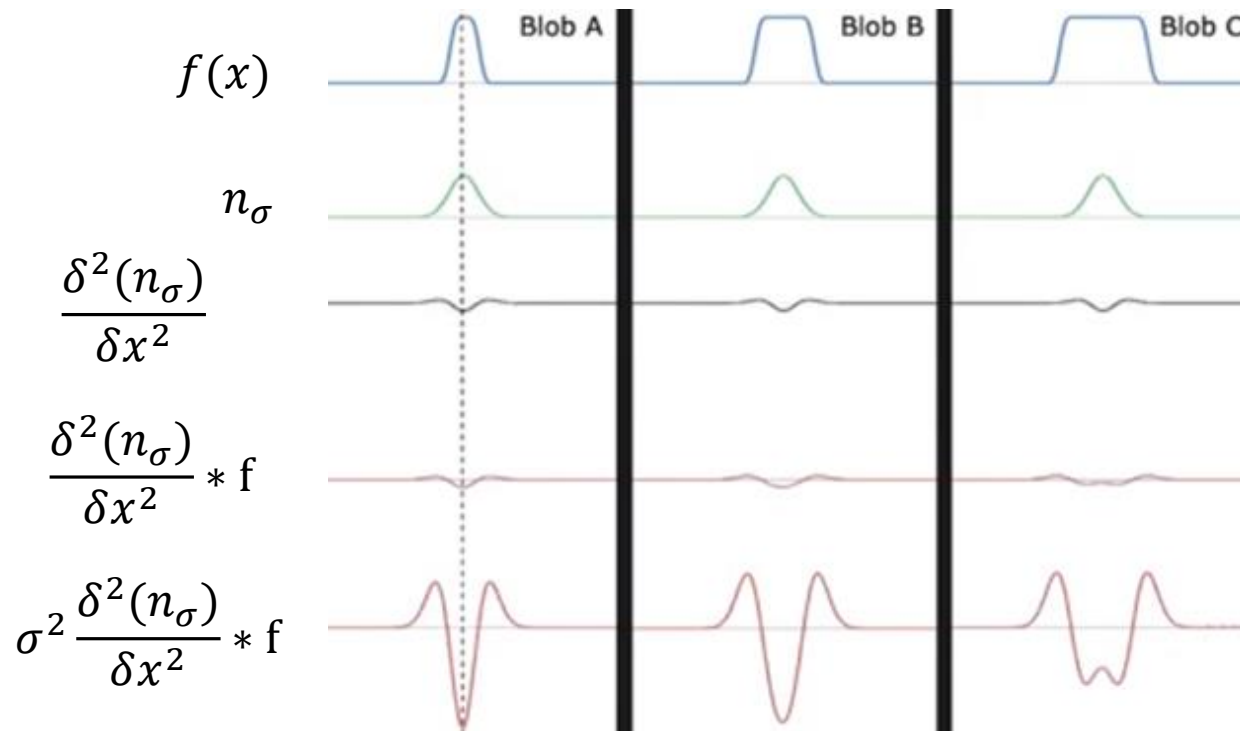
If we can find the pic, we have then found the blob. But, in order to find the two other blobs we increase σ .



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1.3 Detecting blobs

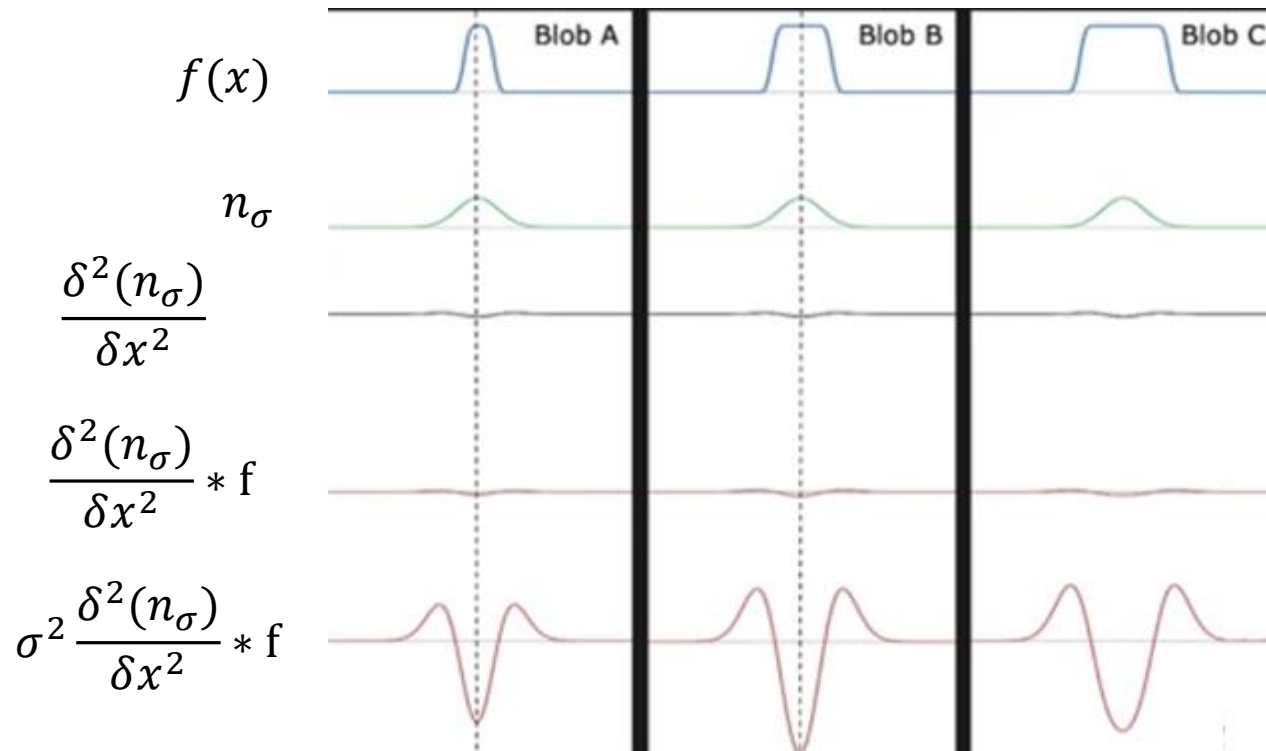
If we can find the pic, we have then found the blob. But, in order to find the two other blobs we continue to increase σ .



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1.3 Detecting blobs

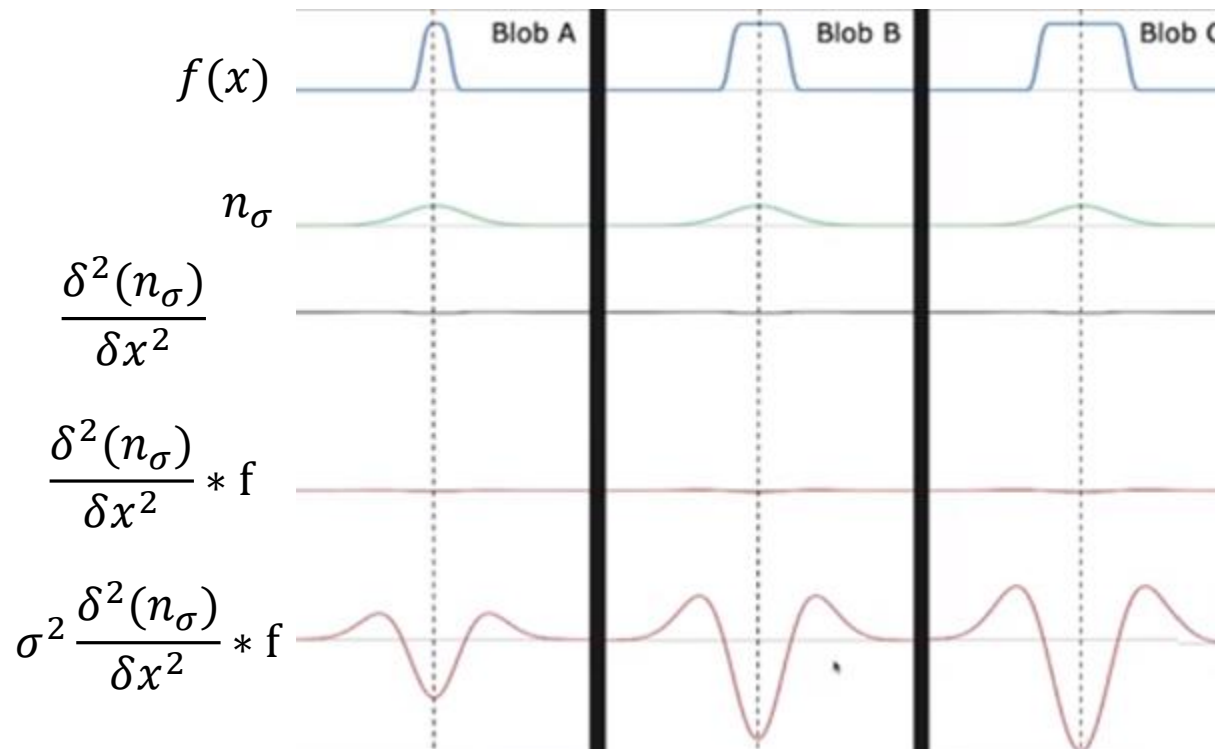
If we can find the pic, we have then found the blob. But, in order to find the two other blobs we continue to increase σ .



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1.3 Detecting blobs

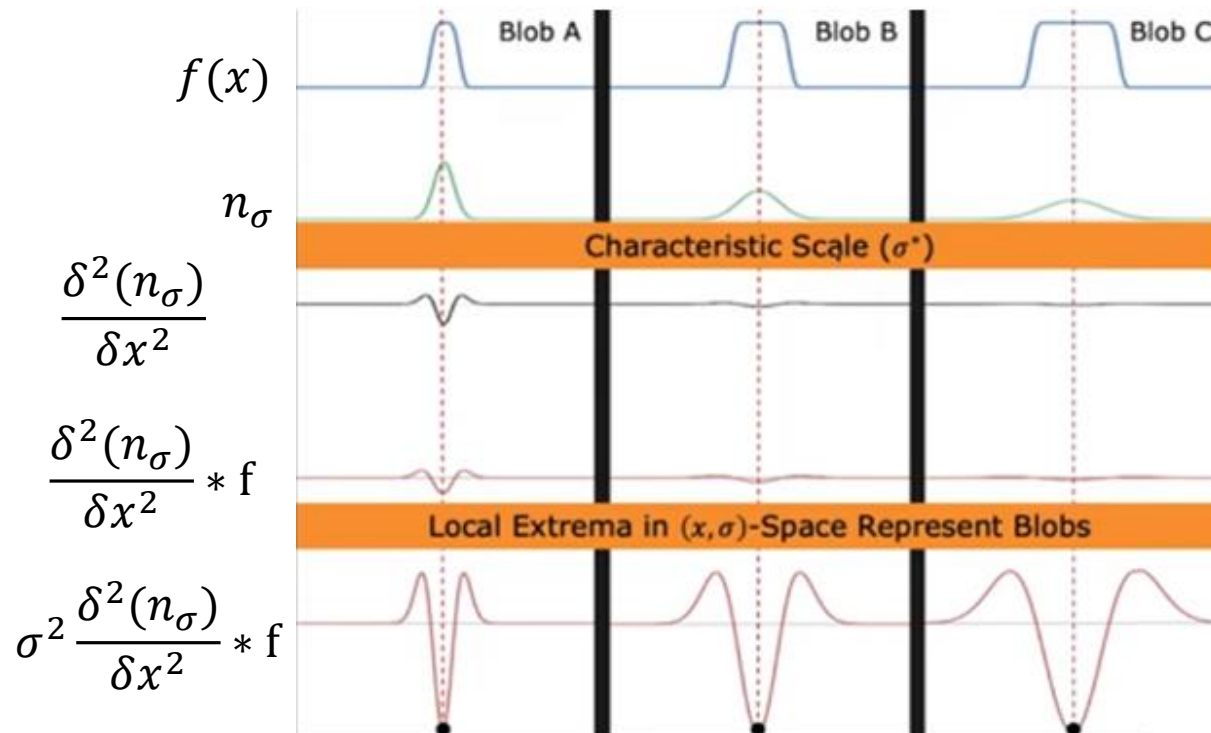
If we can find the pic, we have then found the blob. But, in order to find the two other blobs we continue to increase σ .



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1.3 Detecting blobs

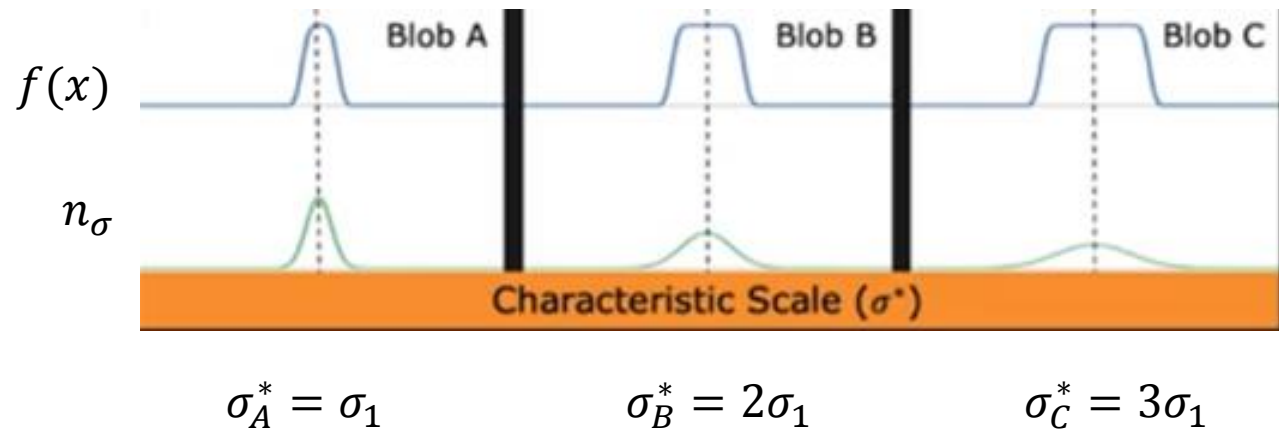
If we can find the pic, we have then found the blob. But, in order to find the two other blobs we continue to increase σ .



Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs

If we can find the pic, we have then found the blob. But, in order to find the two other blobs we continue to increase σ .



Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs

1D Blob Detection Summary

Given 1D signal $f(x)$

Compute $\sigma^2 \frac{\delta^2(n_\sigma)}{\delta x^2} * f$ at many scales $(\sigma_0, \sigma_1, \sigma_2, \dots, \sigma_k)$

Find: $(x^*, \sigma^*) = \arg \max_{(x, \sigma)} \left| \sigma^2 \frac{\delta^2(n_\sigma)}{\delta x^2} * f(x) \right|$

$\left(\begin{array}{l} x^*: \text{Blob position} \\ \sigma^*: \text{Characteristic scale (blob size)} \end{array} \right)$

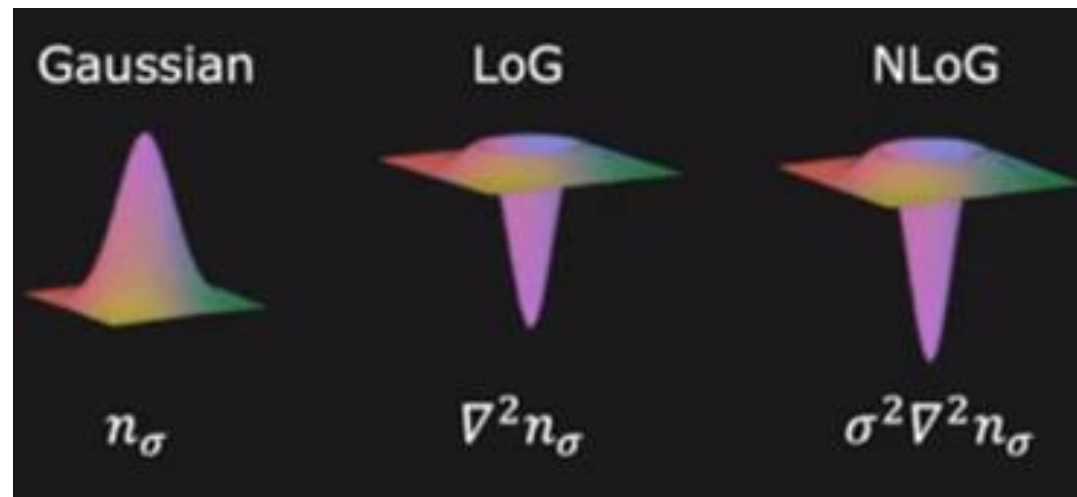
Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs

1D Blob Detection

Normalized Laplacian of Gaussian (NLOG) is used as 2D equivalent for blob detection.

$$\nabla^2 = \frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2}$$



Location of blobs given by local extrema after applying Normalized Laplacian of Gaussian at many scales.

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1.3 Detecting blobs



$S(x, y, \sigma_0)$



$S(x, y, \sigma_1)$



$S(x, y, \sigma_2)$



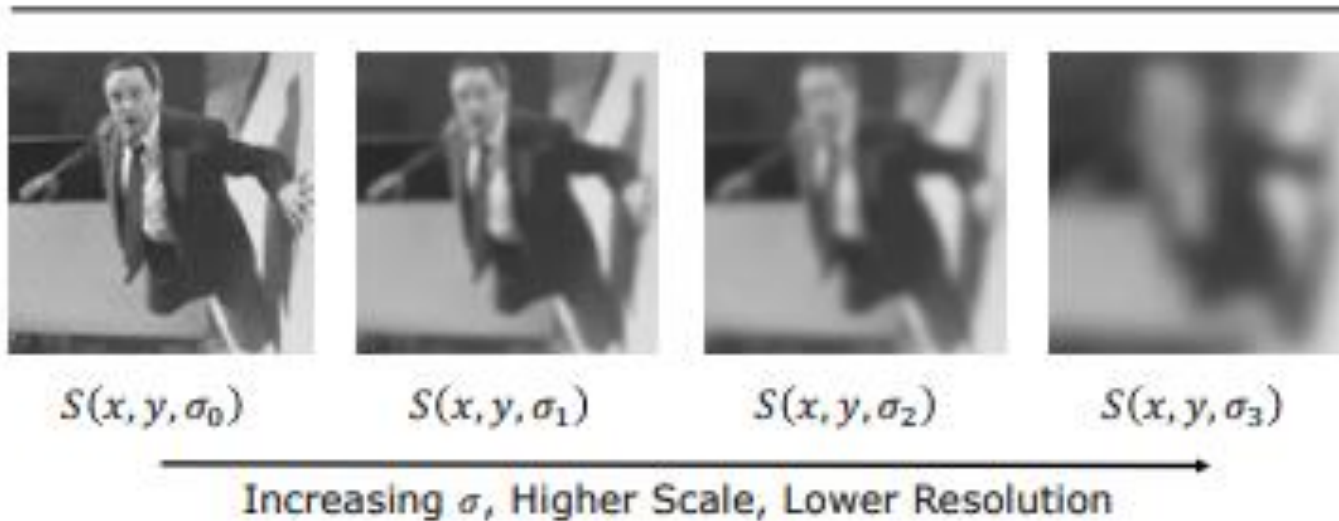
Increasing σ , Higher scale, Low resolution Scale space created by filtering an image with Gaussians of different sigma (σ)

$$S(x, y, \sigma) = n(x, y, \sigma) * f(x, y)$$
$$\sigma^k = \sigma_0 S^k$$

σ_0 : Initial scale
 S : Constant

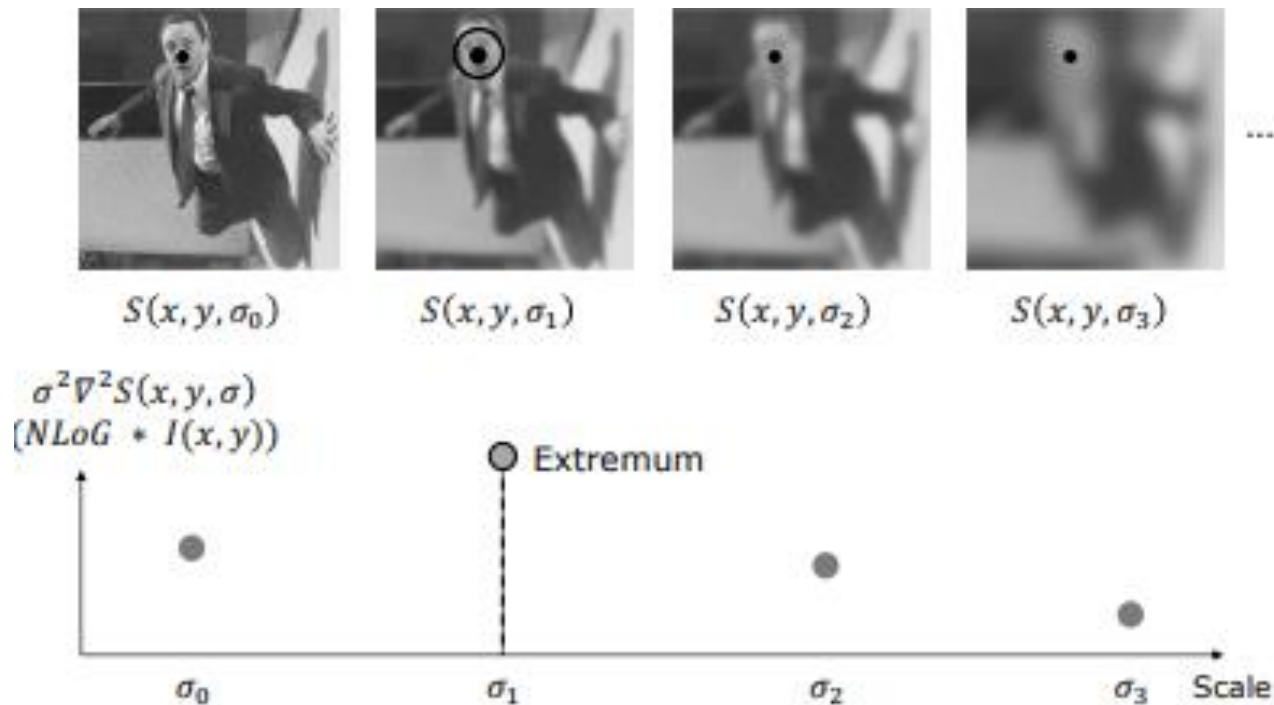
Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs



Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs



The blob is an image patch with uniform brightness and a clear boundary around it.

Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs



Chapitre 5. SIFT Descriptor and Applications

1.3 Detecting blobs

Given an image $f(x,y)$

Convolve the image using NLoG at many scales σ

Find:

$$(x^*, y^*, \sigma^*) = \arg \max_{(x,y,\sigma)} |\sigma^2 \nabla^2 n_\sigma * f(x,y)|$$

(x^*, y^*) : Position of the blob

σ^* : Size of the blob

the values of σ at the extrema correspond to the sizes of the blobs

Chapitre 5. SIFT Descriptor and Applications

1.4 SIFT Detector

Proposed by David G. Lowe.

Distinctive Image Features from Scale-Invariant
Keypoints

International Journal of Computer Vision, 2004

Cited by 65989 papers.

Is widely used in computer vision.



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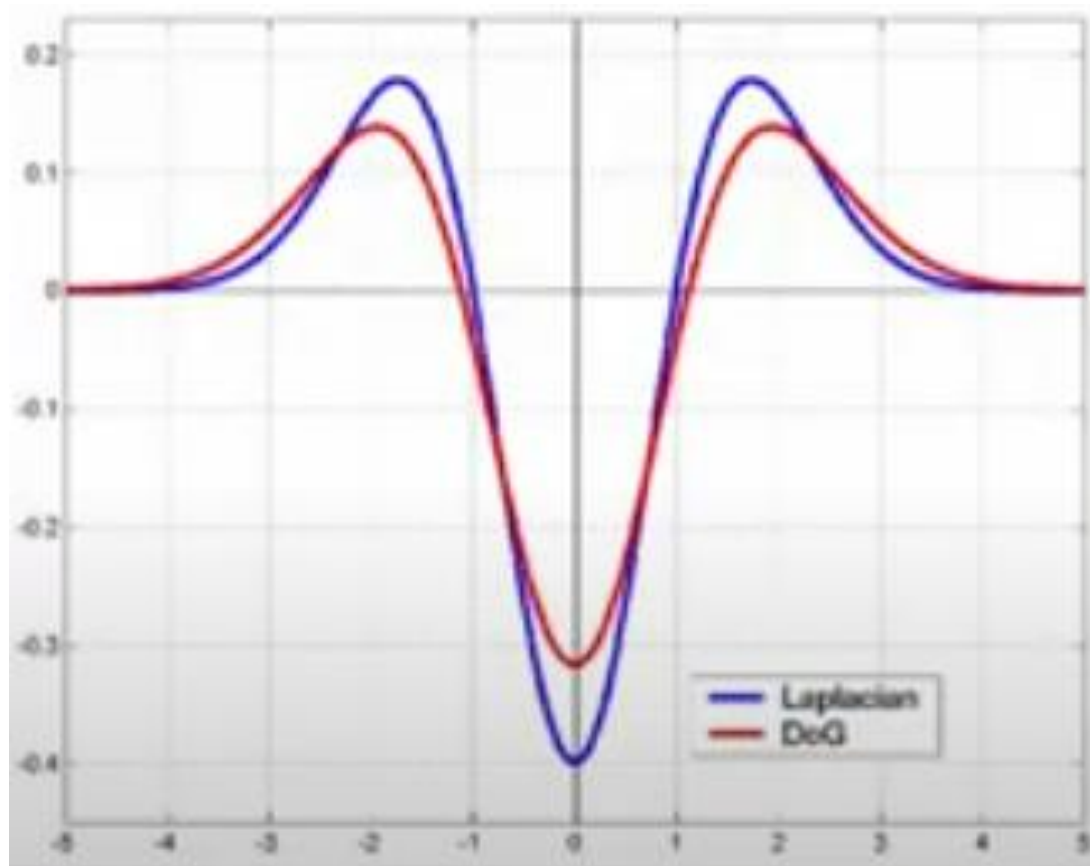
1.4 SIFT Detector

Difference of Gaussian (DOG):

$$DOG = n_{s\sigma} - n_{\sigma} \approx (s - 1)\sigma^2 \nabla^2 n_{\sigma}$$

$$DOG \approx (s - 1)NLOG$$

$$NLOG = \sigma^2 \nabla^2 n_{\sigma}$$



T. Lindeberg, Feature detection with automatic scale selection, IJCV 30(2), pp 77-116, 1998

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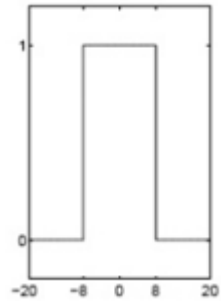
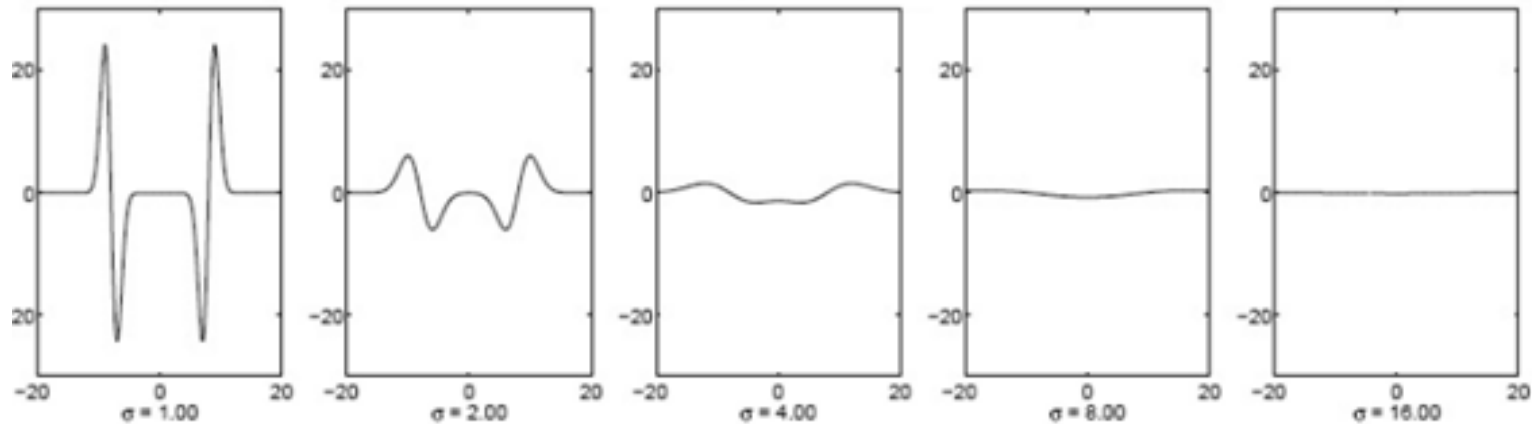
1.4 SIFT Detector



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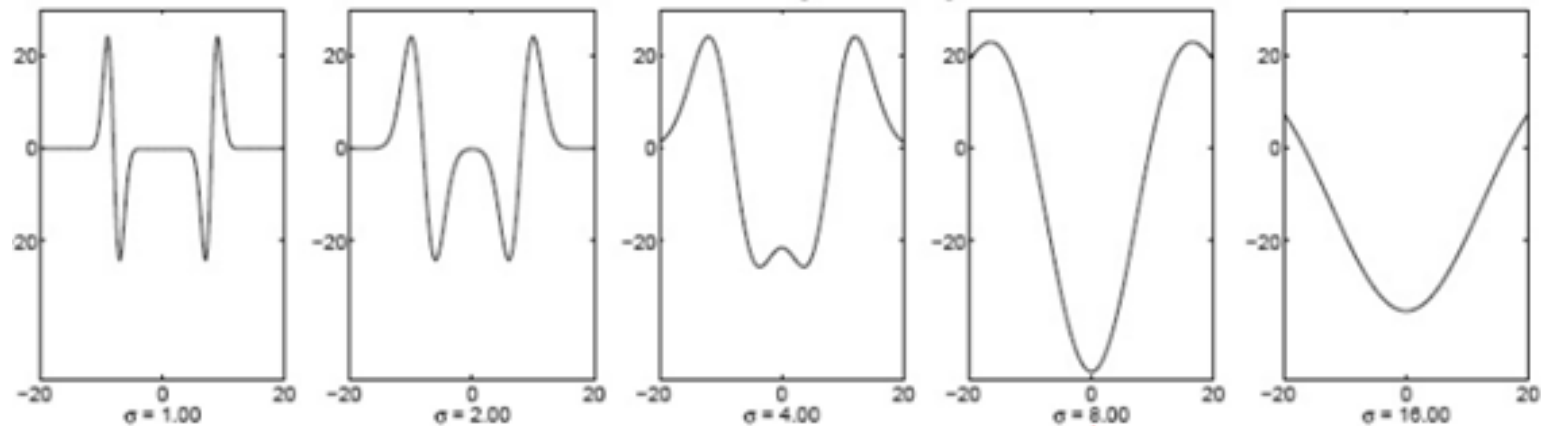
1.4 SIFT Detector

Unnormalized Laplacian response



Original signal

Scale-normalized Laplacian response

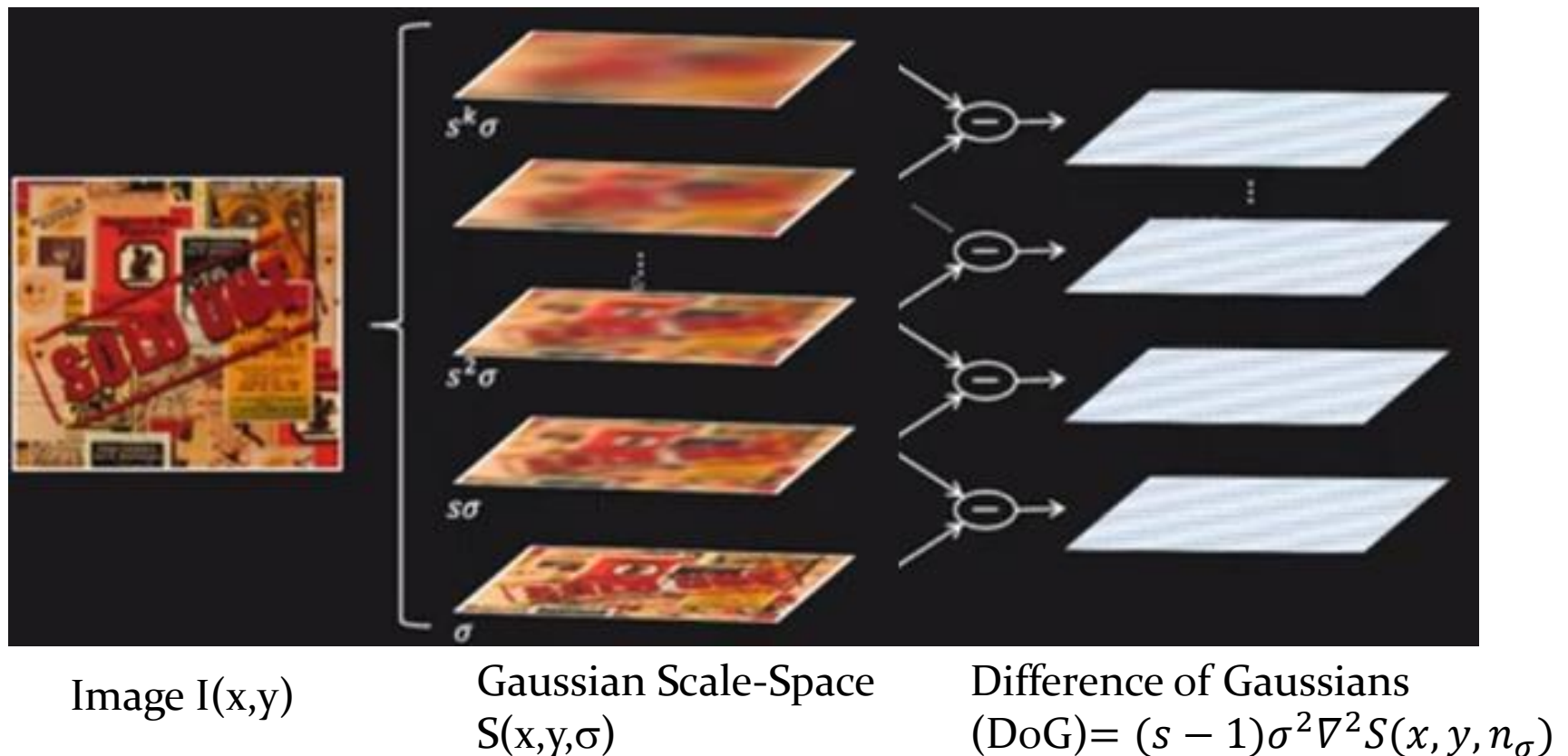


↑
maximum

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1.4 SIFT Detector

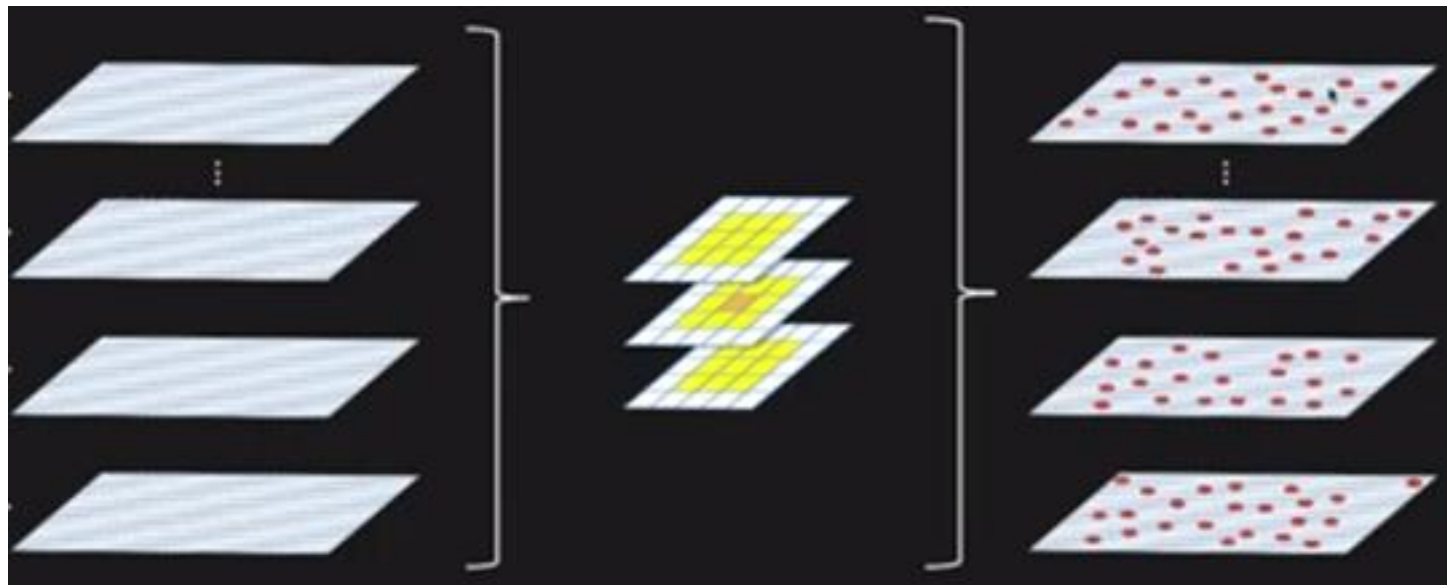
How SIFT is implemented?



Chapitre 5. SIFT Descriptor and Applications

1.4 SIFT Detector

How SIFT is implemented?



Difference of Gaussians
(DoG)

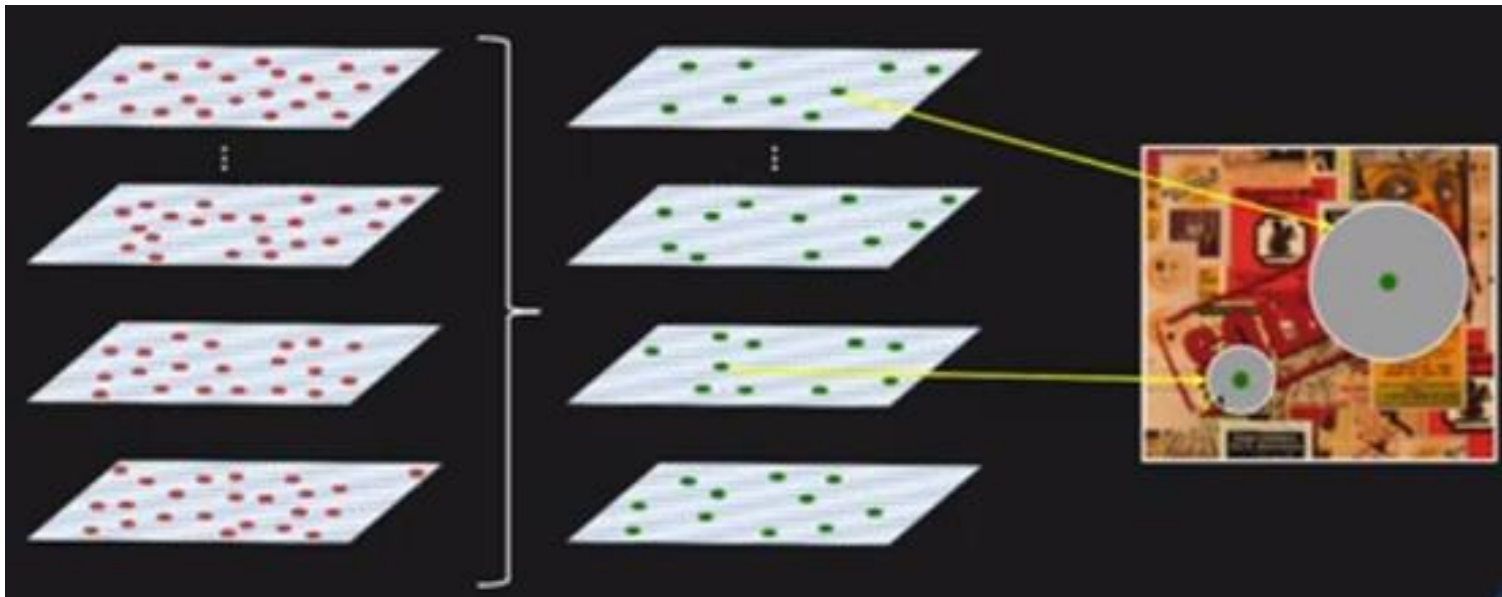
Find Extremum in
every $3 \times 3 \times 3$ grid

Interest point candidates

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1.4 SIFT Detector

How SIFT is implemented?



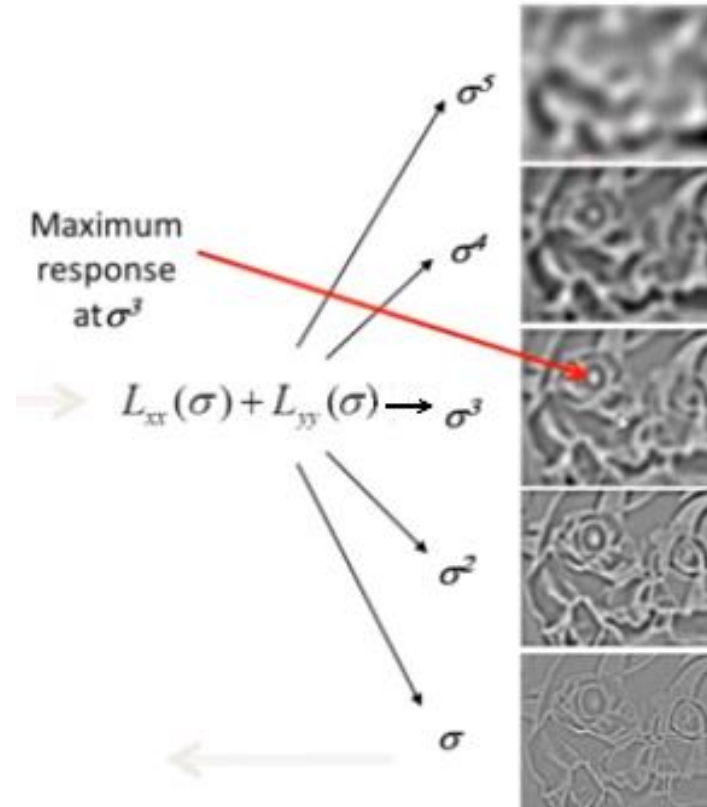
Interest point candidates
(includes weak extrema)

SIFT Interest points
(after removing weak
extrema)

Chapitre 5. SIFT Descriptor and Applications

1.4 SIFT Detector

How SIFT is implemented?



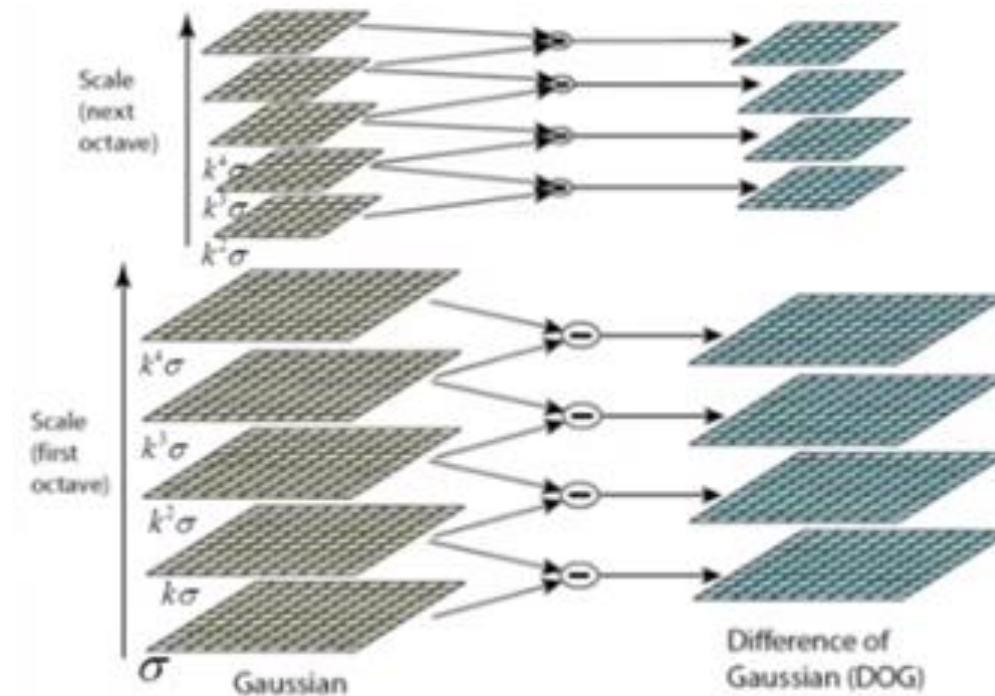
Reference:

<https://www.youtube.com/watch?v=U0wqePj4Mx0>

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1.4 SIFT Detector

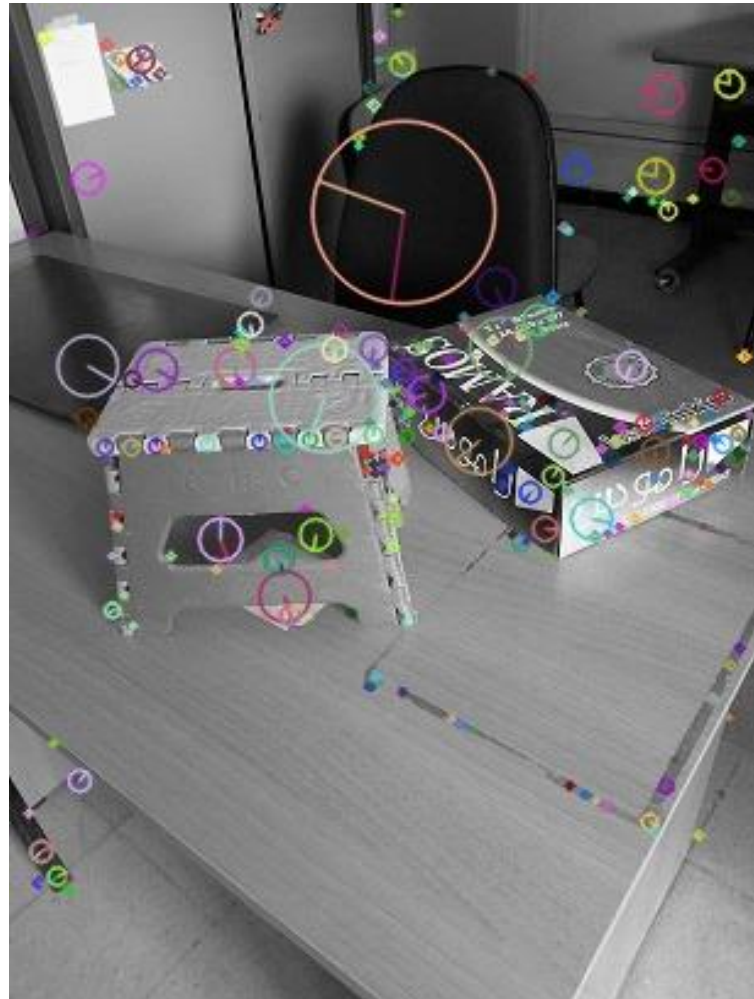
How SIFT is implemented?



Chapitre 5. SIFT Descriptor and Applications

1.4 SIFT Detector

How SIFT is implemented?



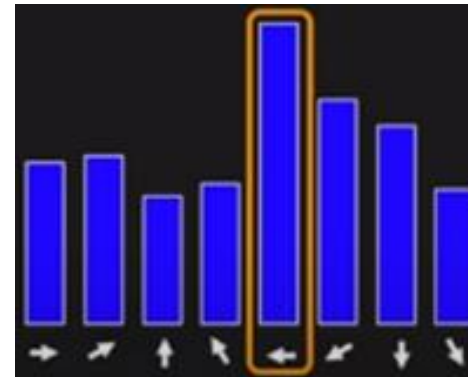
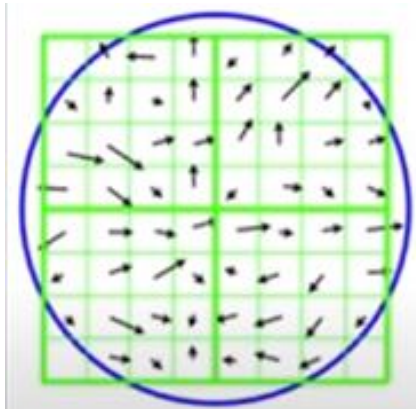
Chapitre 5. SIFT Descriptor and Applications

1.4 SIFT Detector

Use the histogram of gradient directions

Compute the image gradient directions $\theta = \tan^{-1}(\frac{\delta f}{\delta y} / \frac{\delta f}{\delta x})$

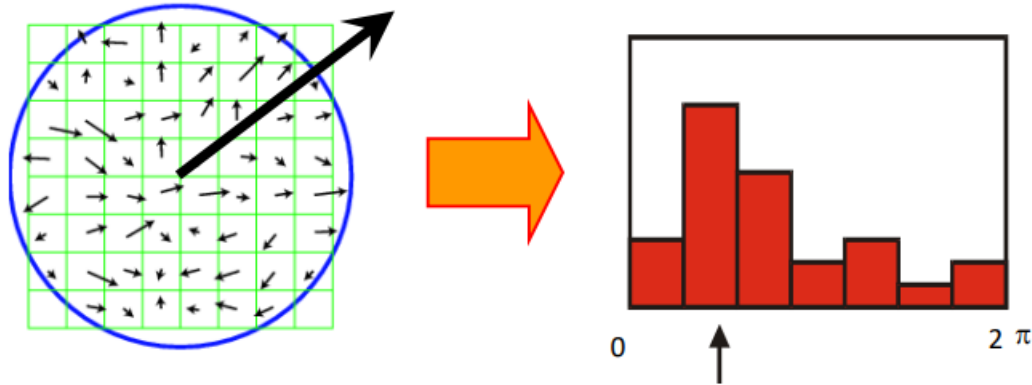
Choose the most prominent gradient direction.



Chapitre 5. SIFT Descriptor and Applications

1.4 SIFT Detector

Assign reference orientation at peak of smoothed histogram

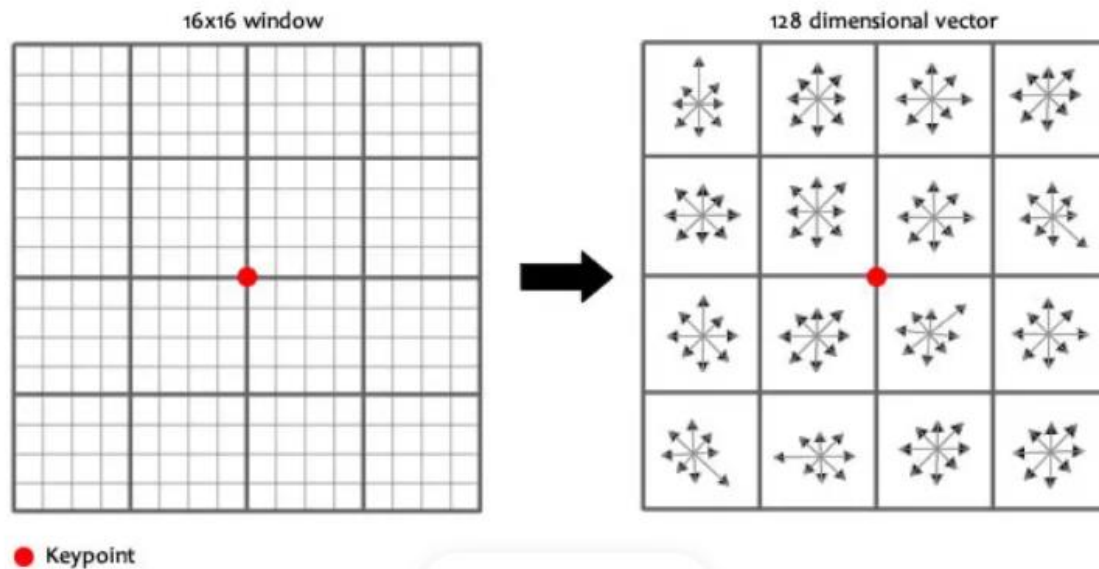


The resulting SIFT descriptor is a length 128 vector representing a 4x4 histogram array with 8 orientation bins per histogram.

Chapitre 5. SIFT Descriptor and Applications

1.4 SIFT Detector

Assign reference orientation at peak of smoothed histogram



The resulting SIFT descriptor is a length 128 vector representing a 4x4 histogram array with 8 orientation bins per histogram.

Chapitre 5. SIFT Descriptor and Applications

1.4 SIFT Detector

SIFT Descriptor

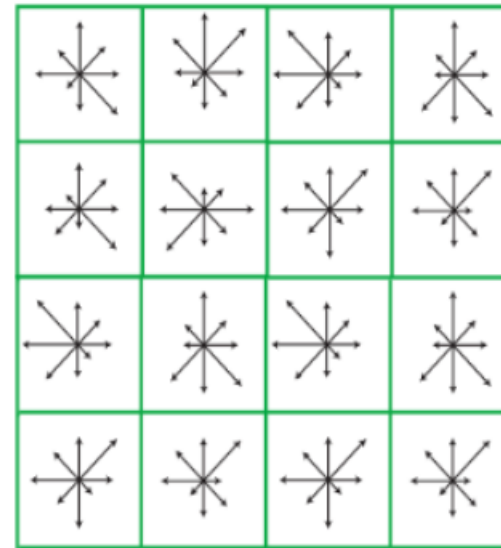
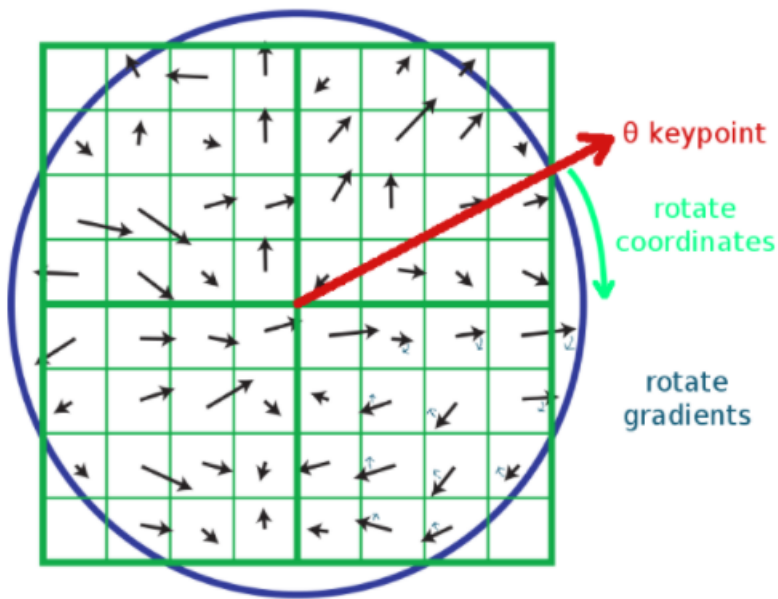
Once the keypoint is found, the next step is to construct a descriptor that contains information of visual characteristics around the keypoint yet is not sensitive to rotation and image illumination.

The steps of building the SIFT descriptor are as following:

1. Use the Gaussian blurred image associated with the key point's scale
2. Take image gradients over a 16×16 array
3. Rotate the gradient directions AND locations relative to the keypoint orientation
4. Create an array of orientation histogram.
5. Add the rotated gradients into their local orientation histograms with 8 orientation bins

Chapitre 5. SIFT Descriptor and Applications

1.4 SIFT Detector

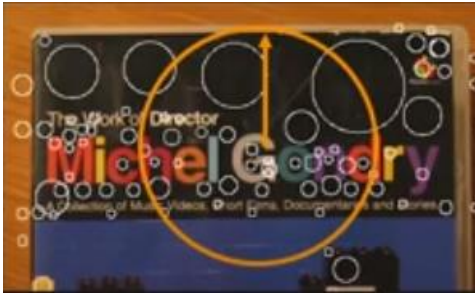


Keypoint descriptor

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1.4 SIFT Detector

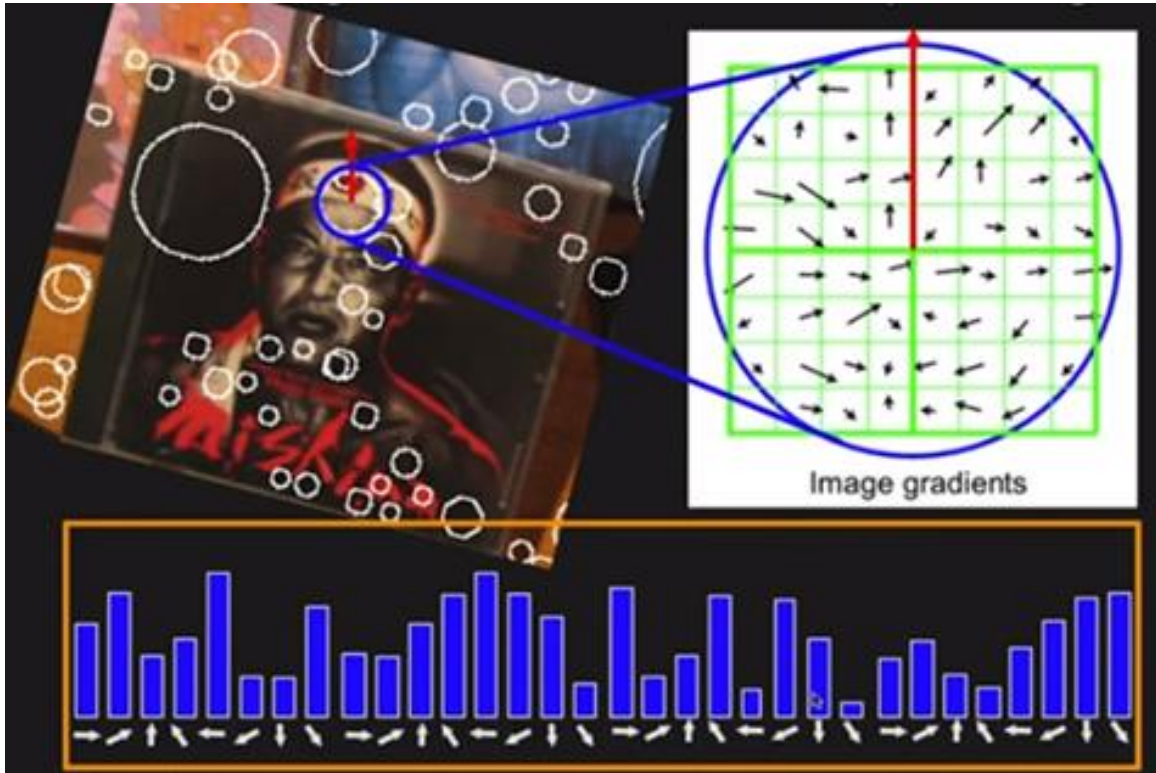
Use the principal orientation to undo rotation



Chapitre 5. SIFT Descriptor and Applications

1.5 SIFT Descriptor

Histogram of gradient directions over spatial directions



Chapitre 5. SIFT Descriptor and Applications

1.5 SIFT Descriptor

Comparing SIFT descriptors

Let $H_1(k), H_2(k)$ be two arrays of data of length N.

L2 Distance:

$$d(H_1, H_2) = \sqrt{\sum_k (H_1(k) - H_2(k))^2}$$

Smaller the distance metric, better the match.

Perfect match when $d(H_1, H_2) = 0$

Chapitre 5. SIFT Descriptor and Applications

1.5 SIFT Descriptor

Comparing SIFT descriptors

Let $H_1(k), H_2(k)$ be two arrays of data of length N .

Normalized Correlation:

$$d(H_1, H_2) = \frac{\sum_k |(H_1(k) - \bar{H}_1)(H_2(k) - \bar{H}_2)|}{\sqrt{\sum_k (H_1(k) - \bar{H}_1)^2} \sqrt{\sum_k (H_2(k) - \bar{H}_2)^2}}$$

Where: $\bar{H}_i = \frac{1}{N} \sum_{k=1}^N H_i(k)$

Larger the distance metric, better the match.

Perfect match when $d(H_1, H_2) = 1$

Chapitre 5. SIFT Descriptor and Applications

1.5 SIFT Descriptor

Comparing SIFT descriptors

Let $H_1(k), H_2(k)$ be two arrays of data of length N .

Intersection:

$$d(H_1, H_2) = \sum_k \min(H_1(k), H_2(k))$$

Larger the distance metric, better the match.